Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026 Topic November 4th

Covariance Matrices from sub-Gaussian Ensembles

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1 Setup

Let $X_1, \dots, X_n \in \mathbb{R}^d$ be independent and identically distributed random vectors with $\mathbb{E}X_i = 0$ and covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$. Define

$$S_n = \frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top. \tag{1}$$

and

$$X = \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \in \mathbb{R}^{n\times d}, \qquad \widehat{\Sigma} = S_n = \frac{1}{n}X^\top X.$$

Because the population mean is zero, $\widehat{\Sigma}$ is the uncentered sample covariance estimator, and $\mathbb{E}\widehat{\Sigma} = \Sigma$. Our goal is to bound the spectral-norm estimation error $\left\|\widehat{\Sigma} - \Sigma\right\|_2$.

For a real matrix A, its largest and smallest singular values are

$$s_{\max}(A) = \sup_{\|x\|_2=1} \|Ax\|_2, \qquad s_{\min}(A) = \inf_{\|x\|_2=1} \|Ax\|_2.$$

2 Section 6.3: sub-Gaussian Random Vectors

Definition 2.1 (Sub-Gaussian random vector)**.** *A zero-mean random vector $X \in \mathbb{R}^d$ is* sub-Gaussian with parameter $\sigma > 0$ *if, for all $\lambda \in \mathbb{R}$ for each fixed $v \in \mathbb{S}^{d-1}$ unit sphere in $\mathbb{R}^d$,*

$$\mathbb{E}\exp\{\lambda\langle v, X\rangle\} \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right). \tag{2}$$

That is, $\langle v, X\rangle$ is a one-dimensional zero-mean sub-Gaussian with parameter σ for any $v \in \mathbb{S}^{d-1}$.

Theorem 2.2. *Suppose $X_1, \dots, X_n \in \mathbb{R}^d$ are i.i.d., mean-zero, and sub-Gaussian with parameter σ. Let $\Sigma = \mathbb{E}[X_i X_i^\top]$ denote the population covariance matrix. The sample covariance $\widehat{\Sigma} = n^{-1}\sum_{i=1}^n X_i X_i^\top$ satisfies the bounds*

$$\mathbb{E}e^{\lambda\left\|\widehat{\Sigma}-\Sigma\right\|_2} \le e^{c_0\frac{\lambda^2\sigma^4}{n}+4d}, \qquad \forall|\lambda| \le \frac{n}{c_1\sigma^2}. \tag{3}$$

Consequently, for all $\delta > 0$,

$$\mathbb{P}\left(\frac{\left\| \widehat{\Sigma} - \Sigma \right\|_2}{\sigma^2} \geq c_2 \left(\sqrt{\frac{d}{n}} + \frac{d}{n} \right) + \delta \right) \leq c_3 e^{-c_4 n \min(\delta, \delta^2)}, \tag{4}$$

and

$$\frac{\mathbb{E}\left\| \widehat{\Sigma} - \Sigma \right\|_2}{\sigma^2} \leq c_5 \left(\sqrt{\frac{d}{n}} + \frac{d}{n} \right). \tag{5}$$

Here $c_0, \ldots, c_5$ are positive universal constants.

Proof. Denote $Q = \widehat{\Sigma} - \Sigma$. Recall from Section 6.1 the variational representation

$$\|Q\|_2 = \max_{v \in S^{d-1}} |\langle v, Qv \rangle| .$$

Let $\mathcal{N} = \{v^1, \ldots, v^N\} \subset S^{d-1}$ be a $\frac{1}{8}$-covering of the unit sphere S^{d-1} with respect to the Euclidean norm. From Chapter 5, there exists such a covering with

$$N \leq \left(1 + \frac{2}{1/8} \right)^d = 17^d. \tag{6}$$

Given any $v \in S^{d-1}$, there exists j such that

$$\left\| v - v^j \right\|_2 \leq \frac{1}{8}.$$

Write $v = v^j + \Delta$, where $\|\Delta\|_2 \leq 1/8$.

Since Q is symmetric,

$$\begin{aligned} v^\top Q v &= (v^j + \Delta)^\top Q (v^j + \Delta) \\ &= (v^j)^\top Q v^j + 2\Delta^\top Q v^j + \Delta^\top Q \Delta. \end{aligned}$$

By Cauchy–Schwarz and the definition of the spectral norm,

$$\begin{aligned} |v^\top Q v| &\leq |(v^j)^\top Q v^j| + 2 \|\Delta\|_2 \|Q\|_2 \left\| v^j \right\|_2 + \|\Delta\|_2^2 \|Q\|_2 \\ &\leq |(v^j)^\top Q v^j| + \left(\frac{1}{4} + \frac{1}{64} \right) \|Q\|_2 \\ &\leq |(v^j)^\top Q v^j| + \frac{1}{2} \|Q\|_2 \\ &\leq \max_{1 \leq j \leq N} |(v^j)^\top Q v^j| + \frac{1}{2} \|Q\|_2 . \end{aligned}$$

Taking the supremum over $v \in S^{d-1}$ gives

$$\|Q\|_2 = \sup_{v \in S^{d-1}} |v^\top Q v| \leq \max_{1 \leq j \leq N} |(v^j)^\top Q v^j| + \frac{1}{2} \|Q\|_2 . \tag{7}$$

Consequently,

$$\|Q\|_2 \le 2 \max_{1\le j\le N} |(v^j)^\top Q v^j|. \tag{8}$$

For all $\lambda \ge 0$,

$$\begin{aligned}\mathbb{E}e^{\lambda\|Q\|_2} &\le \mathbb{E}e^{2\lambda \max_{1\le j\le N} |(v^j)^\top Q v^j|} \\ &\le \sum_{j=1}^{N} \left(\mathbb{E}e^{2\lambda (v^j)^\top Q v^j} + \mathbb{E}e^{-2\lambda (v^j)^\top Q v^j}\right). \end{aligned} \tag{9}$$

Next, we claim that for any fixed unit vector $u \in S^{d-1}$,

$$\mathbb{E}e^{t\langle u, Qu\rangle} \le e^{512 \frac{t^2}{n} e^4 \sigma^4}, \qquad \forall |t| \le \frac{n}{32 e^2 \sigma^2}. \tag{10}$$

We take this bound as given for the moment and use it to complete the proof of (3).

For each vector v^j in the covering set, we apply (10) twice, once with $t = 2\lambda$ and once with $t = -2\lambda$.

Combining the resulting bounds with (9), we obtain

$$\begin{aligned}\mathbb{E}e^{\lambda\|Q\|_2} &\le 2N e^{2048 \frac{\lambda^2}{n} e^4 \sigma^4} \\ &\le e^{c_0 \frac{\lambda^2 \sigma^4}{n} + 4d},\end{aligned}$$

valid for all $0 \le \lambda$, where the last step uses $2(17^d) \le e^{4d}$.

For $\lambda < 0$, the bound follows from $\mathbb{E}e^{\lambda\|Q\|_2} \le 1 \le e^{4d}$.

Having established the MGF bound (3), the tail bound (4) follows from Proposition 2.9.

□

Lemma 2.3. *$\forall u \in S^{d-1}$, ($\langle u, X\rangle$ is 1-dimensional σ-sub-Gaussian),*

$$\mathbb{E}e^{tu^T Q u} \le e^{C\frac{t^2}{n}\sigma^4}, \qquad \forall |t| < \frac{Cn}{\sigma^2}.$$

Proof. Fix $u \in \mathbb{S}^{d-1}$. By definition of Q,

$$\begin{aligned} u^\top Q u &= u^\top \left(\widehat{\Sigma} - \Sigma\right) u \\ &= u^\top \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top - \Sigma\right) u \\ &= \frac{1}{n}\sum_{i=1}^{n} (u^\top X_i)^2 - u^\top \Sigma u \\ &= \frac{1}{n}\sum_{i=1}^{n} \left[(u^\top X_i)^2 - \mathbb{E}(u^\top X_i)^2\right], \end{aligned} \tag{11}$$

since

$$\mathbb{E}(u^\top X_i)^2 = u^\top \mathbb{E}(X_i X_i^\top) u = u^\top \Sigma u.$$

Independence of the observations gives

$$\begin{aligned}\mathbb{E}e^{tu^TQu} &= \mathbb{E}e^{t\frac{1}{n}\sum_{i=1}^n\left[(u^Tx_i)^2-u^T\Sigma u\right]}\\ &= \mathbb{E}\prod_{i=1}^n e^{t\frac{1}{n}\left[(u^Tx_i)^2-u^T\Sigma u\right]}\\ &= \left(\mathbb{E}e^{t\frac{1}{n}\left[(u^Tx_1)^2-u^T\Sigma u\right]}\right)^n.\end{aligned}$$

$\langle u^TX_i\rangle$ is 1-dimensional σ-sub-Gaussian.

Lemma 2.4. *If Y is zero-mean σ-sub-Gaussian, then*

$$Y^2 - \mathbb{E}Y^2$$

is sub-exponential with parameters $(4\sigma^2, 4\sigma^2)$, i.e.,

$$\mathbb{E}e^{t(Y^2-\mathbb{E}Y^2)} \le e^{\frac{t^2}{2}(4\sigma^2)^2}, \qquad \forall|t| < \frac{1}{4\sigma^2}.$$

Proof. Homework. □

By the lemma, $(u^TX_1)^2 - \mathbb{E}(u^TX_1)^2$ is $(4\sigma^2, 4\sigma^2)$-sub-exponential. Therefore,

$$e^{\frac{t}{n}\left[(u^TX_1)^2-u^T\Sigma u\right]} \le e^{\frac{t^2}{2n^2}(4\sigma^2)^2}, \qquad \forall|t| < \frac{n}{4\sigma^2}.$$

□

References

[1] Martin J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019.